MLPR Lab 9

INSTRUCTIONS

- Factory Reports are given as a dataset (factoryReports. xlsx). You need to classify the descriptions given in a dataset as per the category of the description.
- Download and install dependencies/libraries as per you required system preferences.

Step1: Import libraries

- Numpy
- Pandas
- Train test split
- TFID vectorizer from sklearn
- Random Forest classifier
- Accuracy score, confusion matrix
- Matplotlib
- NLTK (Natural Language tool kit- May be need to be installed in your system)
- Stopwords from nltk

Step2: Download "punkt" and "Stopwords" from nltk to do text preprocessing.

Step3: Set a random seed to 0 for reproducibility.

Step4: Load the data file (factoryReports. xlsx) and do preprocessing and removal of noise. Example- Remove all stop words of English language.

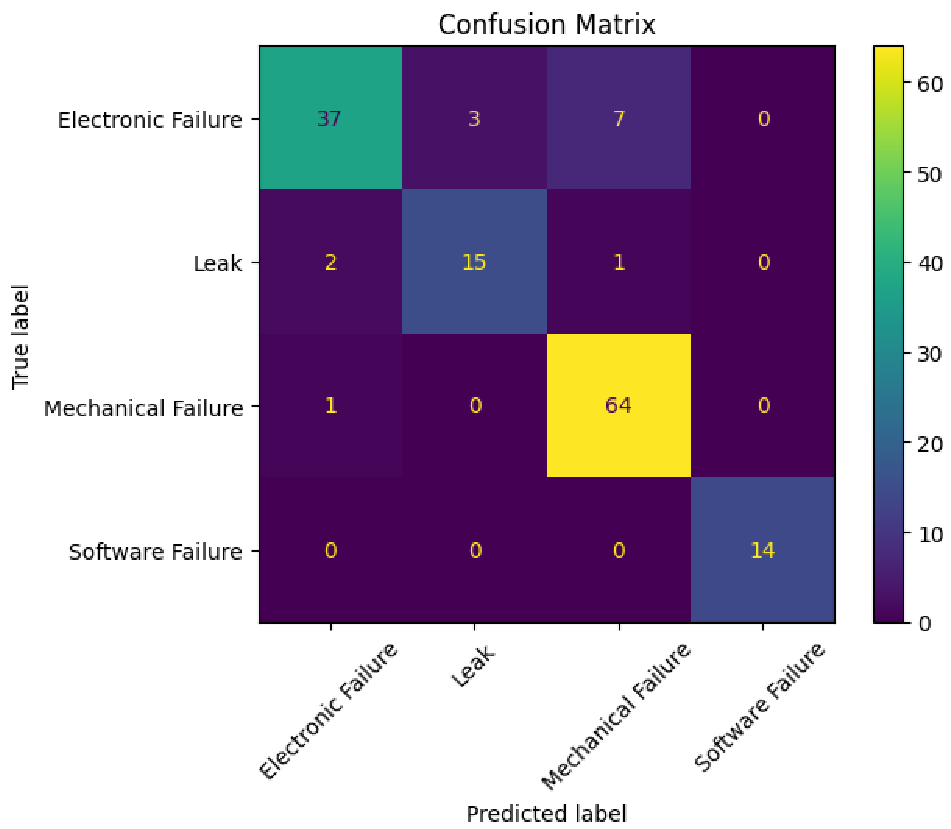
Step5: Tokenize all the individual words in given texts and convert them into lower case for the columns 'Description' and 'Category' columns given in csv file.

Step6: Apply TFIDF vectorizer and then split data into x_train and x_test.

Step7: Train a random forest classifier. Use n_estimators=100 but you can try different values for estimators and choose the parameter that gives you optimal performance. Use random_state=42.

Step8: Fit the model and make prediction.

Step9: Calculate and print the accuracy. Also plot the confusion matrix shown below as an output reference



<u>Report: </u> Answer the following 5 questions within your report:

1. Which techniques would you use for keyword normalization in NLP, the process of converting a keyword into its base form?
2. What is TF-IDF in NLP?
3. What are some alternatives for TF-IDF method to parse text data?
4. What advantage(s) does word2vec has over TF-IDF?
5. Provide two applications of the above lab assignment problem.

<u>DELIVERABLES:</u> <u>Items to be uploaded on LMS-</u>

- Main code file along with the outputs, the image plots shown above and the report answers in one file in PDF format
- **Please note: Write your comments along with the code for each step as required. All outputs and plots must have correct labels, titles, axes titles etc.**
- Report answers must be written in **full and complete sentences**. Answer each question in a new line.
- Your file should be as follows. **Yourname_lab9**.pdf
- Due time and date are given on LMS. Submit it before the deadline.